

Dillon Bowes

<https://www.linkedin.com/in/dillon-bowes> | 858-401-9297 | dillon.bowes0@gmail.com | dillonbowes.github.io

PROFESSIONAL SUMMARY

Published new professional with solid understanding and application of consulting with applied experience across AI governance, cybersecurity, defense R&D, financial modeling, and econometric research. Background includes designing and testing computational systems for autonomous platforms, publishing cybersecurity and AI maturity research, and developing novel coding algorithms in both education and industry contexts. Proficient in Python-based analysis, statistical modeling, and machine learning, with a focus on differential equations and vector calculus.

EDUCATION

University of California San Diego

September 2022-June 2026

Bachelor of Science in Economics, Minor in Data Science (Halicioğlu Institute for Data Science)

TECHNICAL SKILLS

Languages: Python, R, Ruby/HTML, CSS, Java

Packages/Visualization: SciPy, Matplotlib, NumPy, Pytorch, Pandas, ggplot2, plotly, seaborn

Machine Learning & AI: PyTorch, AI governance consulting, enterprise AI readiness assessment, AI maturity modeling, operational risk analysis

Finance and Economic Modeling: Financial analysis, economic modeling, statistical forecasting, risk analysis, Excel based modeling

EXPERIENCE

Fire Mountain Labs – Remote/San Diego, CA

July 2024 – Present

AI Risk & Governance Consulting Intern

- Published author of a peer-reviewed paper “Operationalizing AI Assurance: Translating Maturity and Risk Frameworks into Measurable Practice” in the Society Photo-optical Instrumentation Engineers (SPIE) that addressed operational gaps between governance strategy, risk assessment, and enterprise implementation.
- As lead technical evaluator for a customer facing AI solution under consideration, identified significant defects, reliability problems and cyber vulnerabilities that resulted in a quick executive decision and avoided risk to the company. CEO: “Working with Dillon has already been a highlight, he’s bringing sharp thinking to the hard questions around AI maturity and risk,”
- Applied cyber defense methodologies like NIST, OWASP, and MITRE ATLAS to accurately assess AI readiness resulting in full restructuring of multiple firms cyber strategies and setups.

Economics Research Lab – Dr. Alisher Batmanov – UCSD

September 2023 –January 2025

Research Assistant (Python)

- Contributed to a research paper titled Mental Models, Social Learning, and Statistical Discrimination by implementing refined reinforcement learning models and statistical analyses of CDFs for experimental applications.
- Utilized Python extensively for data scraping, cleaning, and analysis to support the study of large-scale behavioral science.

- Investigated the role of psychological factors versus financial incentives in hiring decisions, through Pandas and SciPy analyzing how education and other candidate characteristics influence hiring outcomes.

Brains and Motion Education – San Diego, CA

June 2024 – June 2025

Instructor, Instruction/Curriculum Development

- Taught coding to children and teens as well as created lesson plans, focusing on HTML and a custom-designed introductory coding language.

UCSD– San Diego, CA

September 2022 – January 2025

Mentor, Mentor Collective Program

- Mentored and tutored three freshmen at UCSD on class schedules, faculty engagement, professional development, and relevant classes in data science and statistics.

NIWIC PAC– Point Loma, CA

June 2022– September 2022

Data Analysis Intern, Autonomous Systems

- Developed novel data analysis and computational fluid dynamics programs to accurately test submersible vehicles.
- Delivered, modeled, and tested aquatic drones in Inventor CAD and Fusion, using the aforementioned data and programs.

RELEVANT COURSEWORK

Mathematical Foundations of Computer Science; Regression Discontinuity Designs, Casual Inference Methods in Econometrics, Probabilistic Modeling, Machine Learning, Advanced Econometrics, Data Mining, Decisions Under Uncertainty, Applied Linear Algebra for Machine Learning, Data Analysis and Inference, Calculus and Analytical Geometry for Science and Engineering.

PROJECTS AND PAPERS

Operationalizing AI Assurance: Translating Maturity and Risk Frameworks into Measurable Practice

- AI assurance frameworks; MITRE/NIST maturity models; AI security benchmarking; Python analysis; governance and risk operationalization; industry consultation

The Economics of Wartime Trading

- Wartime trading analysis; market dynamics; volatility modeling; investor behavior; economic uncertainty

Independent Investment & Black Box Modeling

- FX modeling; quantitative trading; risk modeling; portfolio optimization; automated decision systems

Personal Professional Website

- Web development; HTML/CSS/JavaScript; UI/UX design; interactive portfolio implementation

Company Financial & Strategic Analysis – Research Paper

- Financial modeling; Python analytics; scenario modeling; company valuation; strategic and market analysis

ACTIVITIES

2018 - Present Volunteer, Ronald McDonald House

2022 - Present Member, Alzheimer's Association